# AIN SHAMS UNIVERSITY

Faculty of Engineering

Mechatronics Engineering Department


CSE480: Machine Vision

# Handwritten Digit Classification

Lab Assignment #06

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| **Subject:** | K-Nearest Neighbors Digit Classification |
| **Dataset:** | Scikit-Learn Digits Dataset |
| **Implementation:** | Python, Scikit-Learn, Matplotlib |
| **Model:** | KNN Classifier (k = 3) |
| **Semester:** | Spring 2026 |
| **Prepared by:** | Mohammed Mahmoud AlJamal 2002612 |

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# 1 Introduction

This lab demonstrates a supervised machine learning pipeline using the Scikit-Learn handwritten digits dataset. The objective is to classify handwritten digits using the K-Nearest Neighbors (KNN) algorithm.

The workflow includes:

- Loading the digits dataset
- Visualizing handwritten digit images
- Splitting the dataset into training and testing sets
- Standardizing the feature vectors
- Training a KNN classifier
- Evaluating classification performance
- Generating visual analysis plots

# 2 Dataset Description

The dataset used is the built-in load_digits() dataset from Scikit-Learn.

Dataset properties:

- Total samples: 1797
- Image size: $8 \times 8$
- Number of classes: 10 digits (0–9)
- Feature vector size: 64 features

# 3 Methodology

## 3.1 Train-Test Split

The dataset was divided into training and testing sets using:

- Test size = 0.25
- Random state = 42

## 3.2 Standardization

Feature scaling was performed using StandardScaler() to normalize the data before training the KNN classifier.

## 3.3 KNN Classifier

The classifier used was:

- Algorithm: K-Nearest Neighbors
- Number of neighbors: $k = 3$

# 4 Results

## 4.1 First 50 Digits Visualization


Figure 1: Visualization of the first 50 handwritten digit samples

## 4.2 Confusion Matrix

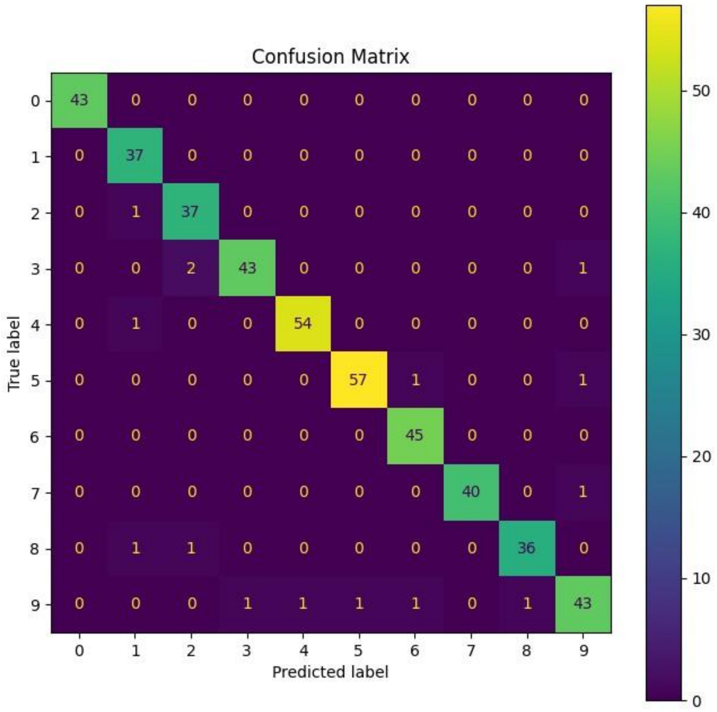


Figure 2: Confusion matrix of KNN classification results

## 4.3 Sample Predictions

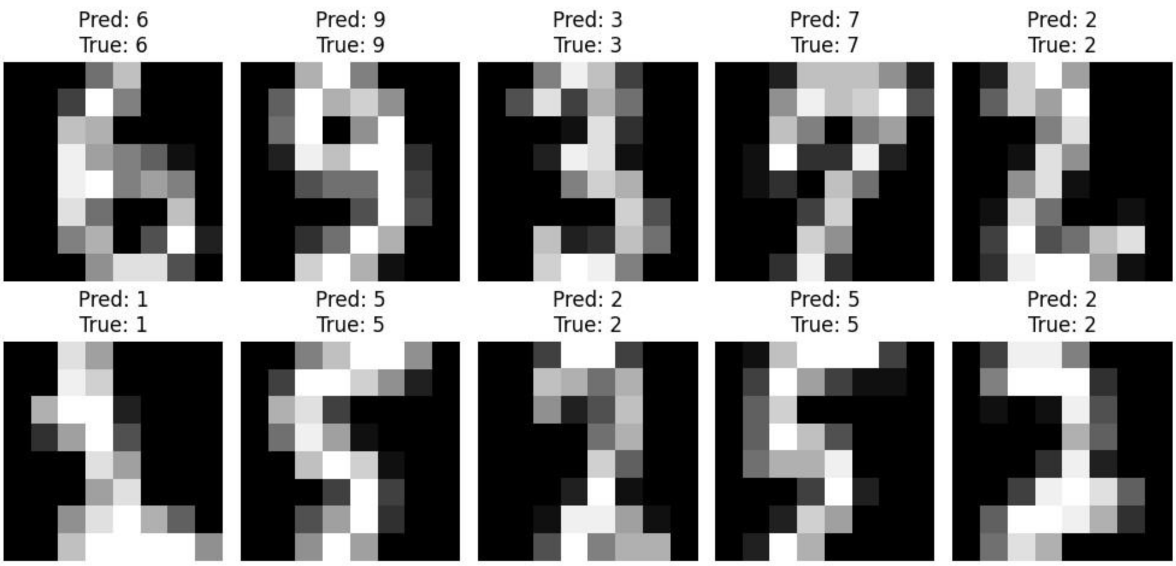


Figure 3: Example predictions from the testing dataset

### 4.4 Accuracy Graph

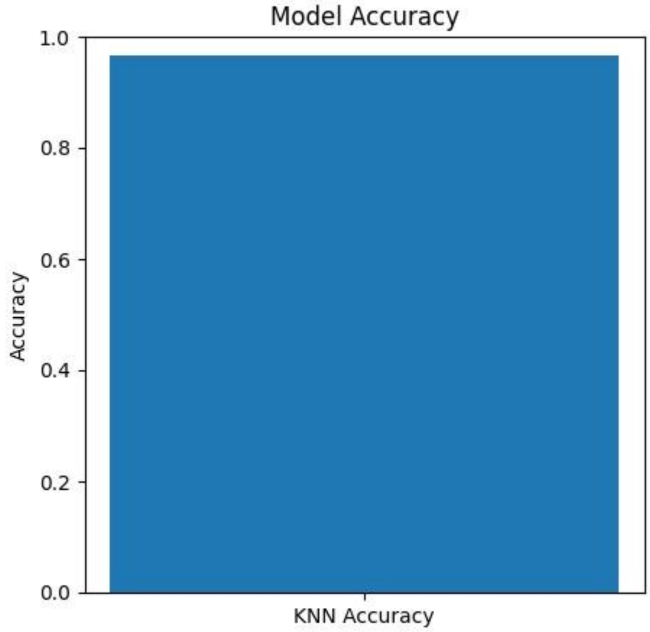


Figure 4: Model accuracy visualization

## 5 Performance Evaluation

The KNN classifier achieved very high classification accuracy on the test dataset, demonstrating the effectiveness of KNN for small image classification tasks.

The confusion matrix shows that most predictions lie on the main diagonal, indicating correct classifications with very few misclassified samples.

# 6 Conclusion

In this lab, a complete handwritten digit classification pipeline was implemented using Python and Scikit-Learn.

The implemented system successfully:

- Loaded and visualized digit images
- Performed preprocessing and standardization
- Trained a KNN classifier
- Evaluated model performance using multiple visualization tools

The achieved results demonstrate that KNN is highly effective for the classification of low-dimensional handwritten digit datasets.

GitHub Link for the Lab: MohammedAlJamal10/lab5: K-Nearest Neighbors Digit Classification